

Replication Guide

This document provides step-by-step instructions to fully replicate the experimental results reported in the accompanying paper.

1. Prerequisites

Ensure all dependencies listed in **requirements.txt** are installed before proceeding.

2. Clone the Repository

```
git clone https://github.com/YOUR_USERNAME/bug-report-classification.git
cd bug-report-classification
```

3. Obtain the Datasets (files already placed, just for demonstration)

Download the five CSV files from the official course repository:

```
https://github.com/ideas-labo/ISE/tree/main/lab1
```

Place all five CSV files into the **data/** subfolder (files already placed, just for demonstration):

```
data/pytorch.csv
data/tensorflow.csv
data/keras.csv
data/incubator-mxnet.csv
data/caffe.csv
```

4. Install Dependencies

```
pip install pandas numpy scikit-learn xgboost scipy nltk matplotlib seaborn
```

5. Run the Main Experiment Script

```
python classify.py
```

Expected runtime: approximately 5–15 minutes depending on hardware (tested on Apple M2, 8 GB

RAM).

6. Expected Outputs

After completion, the following files will be generated:

- **results/raw_results.csv** — Raw per-repeat values for NB, XGB-TF-IDF (ablation), and XGBoost (proposed).
- **results/summary_table.csv** — Mean $\pm$ std summary table.
- **figures/fig1_metric_comparison.png**
- **figures/fig2_f1_boxplot.png**
- **figures/fig3_training_time.png**

7. Verifying Results

The terminal output will print mean and standard deviation for all three models (NB, XGB-TF-IDF, XGBoost Proposed). These values should match Table 1 in the report within ± 0.002 (minor floating point variance is expected across platforms).

8. Stochastic Reproducibility

The experiments use fixed random seeds (seed 0 through 29 for the 30 repeats, and random_state=999 for the initial shuffle). These are hardcoded in `classify.py`, ensuring full reproducibility.

9. Raw Results Data

All raw result CSVs are also committed to the repository under the **results/** folder so that results can be inspected without re-running the experiments.